

WARNINGS ABOUT SIMULATION REVISITED: IMPROVING OPERATIONS IN CONGONHAS AIRPORT

FABIO VITOR

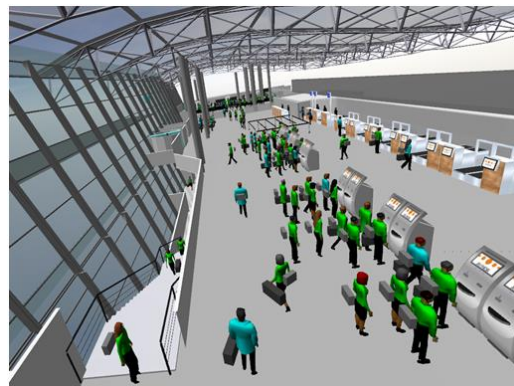
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DDV Credit Advisory / Maua Institute of Technology

WHAT IS THE GOAL OF THIS STUDY?

- Many academics and practitioners have recently discussed warnings about simulation that can lead to the failure of a simulation project.
- This study summarizes many of the concerns described by Banks and Chwif (2010) from an application perspective.
- The study evaluates the check-in area of one of the most important airports in Brazil (Congonhas Airport) using discrete event simulation.
- The paper highlights the contribution of many of the pitfalls described by Banks and Chwif (2010) to improve the check-in of Congonhas Airport.



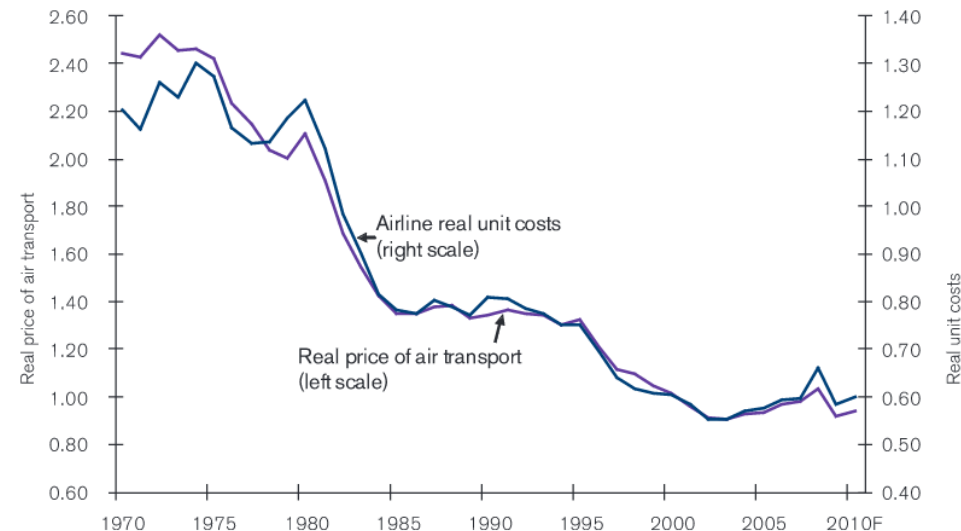
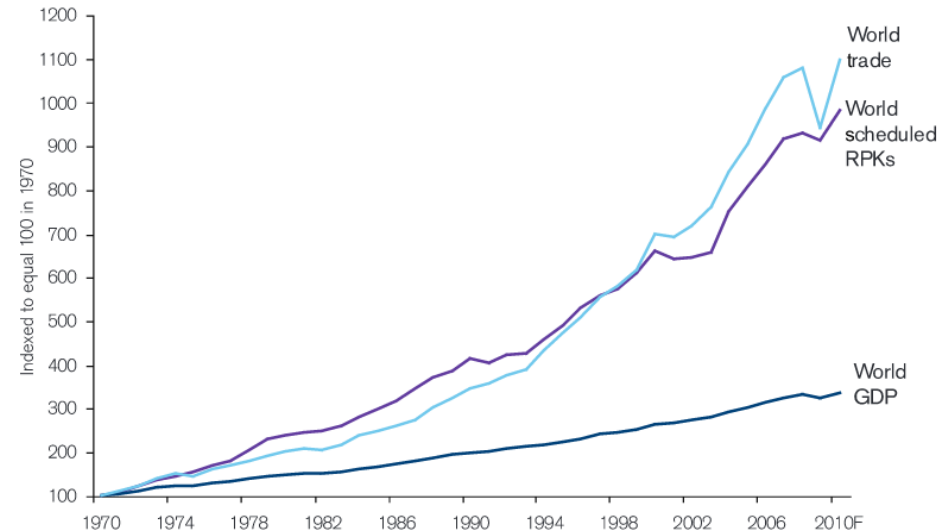
Source: Airport Research Center GmbH

WARNINGS ABOUT SIMULATION

- Banks and Chwif (2010) divide the warnings in seven categories:
 - **Data Collection;**
Quality/Amount of Input Data, Statistical Distribution, Breakdowns, Data Acquisition
 - **Model Building;**
Construct Proper Model, Conceptual Model prior Implementation, Gradual Validation
 - **Verification and Validation;**
Perform a lot of Verification and Validation
 - **Analysis;**
Do not Simulate Output Data, Avoid Point Estimates, Warm-Up, Performance Measures
 - **Simulation Graphics;**
Organize Models Properly instead of Fancy Graphics
 - **Management of the Simulation Process;**
Principles from Project Management, Objective of the Study, No Overestimation, Follow all Phases
 - **Human Factors, Knowledge, and Abilities.**
Statistics, Spreadsheet Skills, “Humanware”

AIRLINE INDUSTRY WORLDWIDE

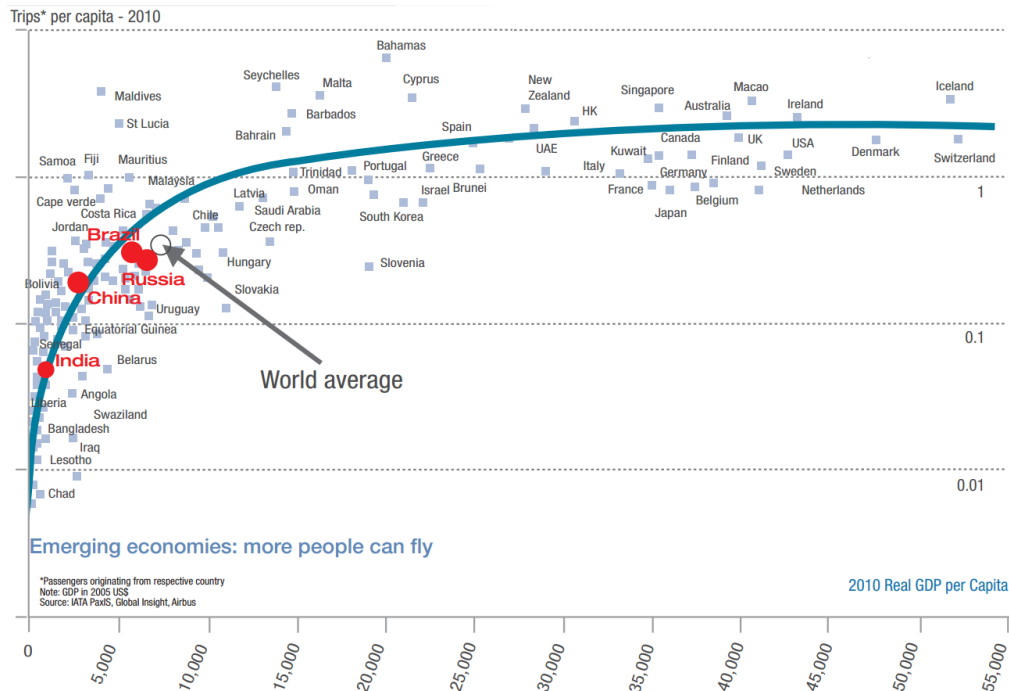
- Revenue Passenger Kilometers (RPK) increased more than 10 times since 1970.
- 43% of passengers travel between USA, Canada, Western Europe and Japan; but will be only 30% in 2030.
- Airline Industry in Latin America will increase 7.2% per year until 2030, just 0.3% behind China.
- The cost of air transportation and airfare decreased more than 60% since 1970.



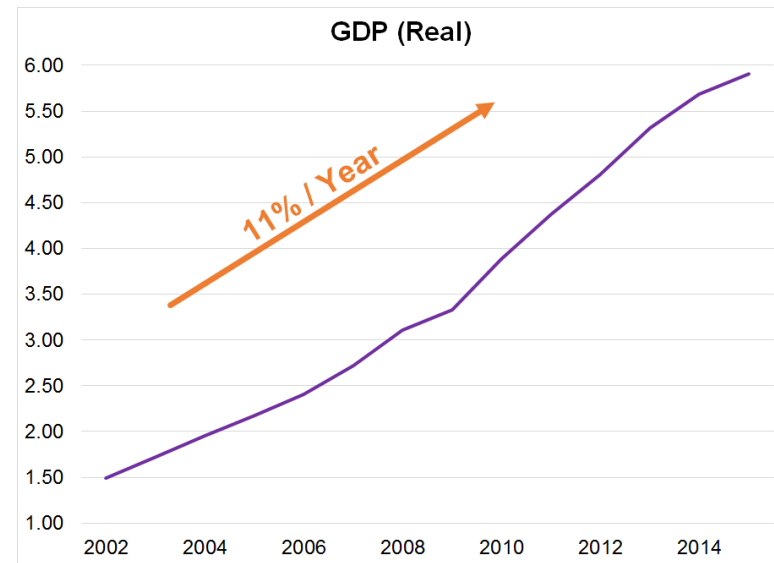
Source: IATA (2011) – Vision 2050

AIRLINE INDUSTRY IN BRAZIL

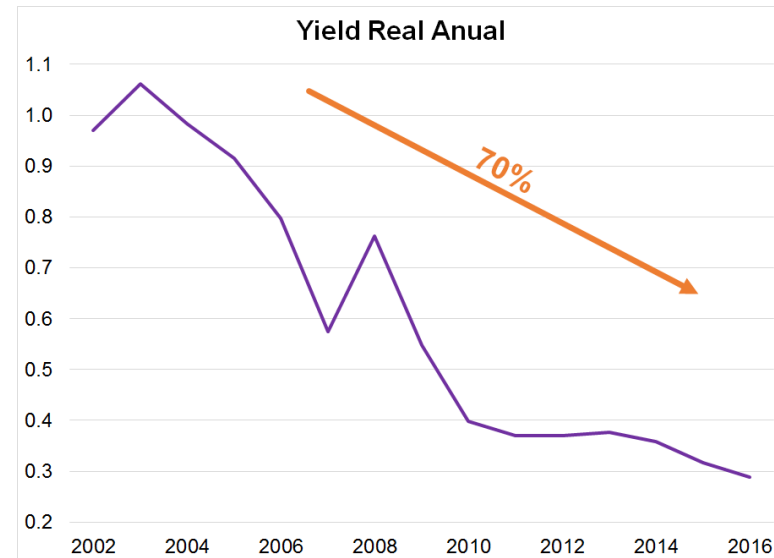
- Gross Domestic Product (GDP) in Brazil increased on average 11% per year from 2002 to 2015.
- Airfare in Brazil decreased more than 70% from 2002 to 2016.



Source: Airbus (2011)



Source: IBGE (2016)

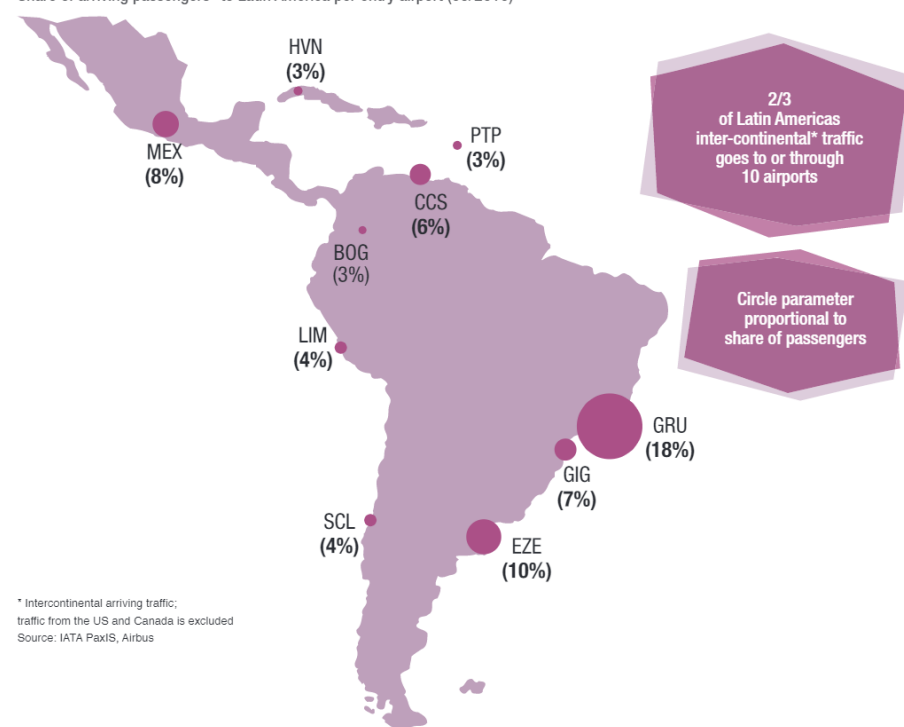


Source: ANAC (2016)

AIRLINE INDUSTRY IN BRAZIL

- Domestic flight traffic in Brazil is expected to be the 15th largest in the world by 2030, increasing at a rate of 6.6%.
- 25% of total international travel in Latin America occurs in two majors airports of Brazil: the International Airport of São Paulo (GRU) and the International Airport of Rio de Janeiro (GIG).
- 9.1% of the GDP in Brazil originates from tourism and business travel, which generate approximately 8 million jobs.

Share of arriving passengers* to Latin America per entry airport (09/2010)



Source: Airbus (2011)

AIRPORTS IN SÃO PAULO

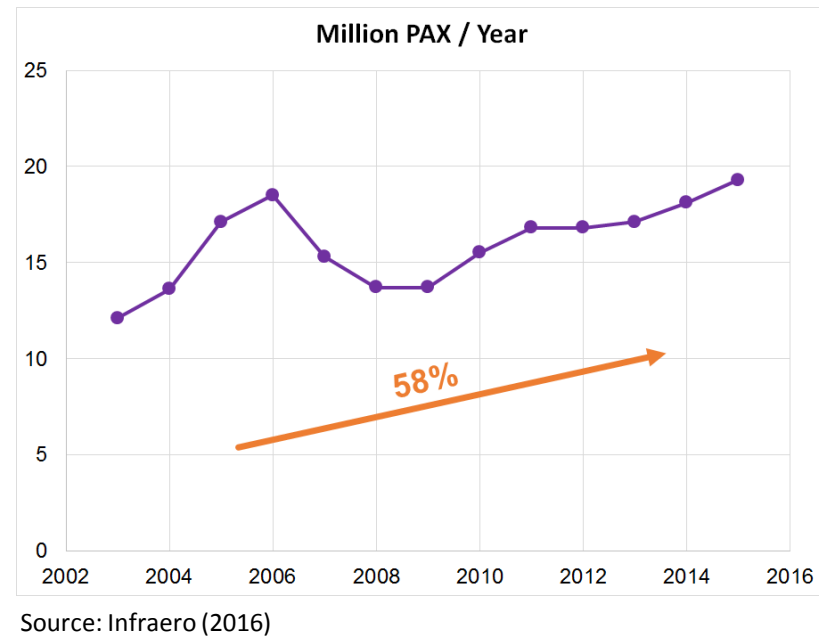
- São Paulo is the hub of air transportation in Brazil because it has the highest GDP of Brazil and Latin America. Moreover, São Paulo is expected to have the 6th highest GDP of the world by 2025. Three major airports in Brazil are located in São Paulo.
- International Airport of São Paulo (GRU).
39 million passengers in 2015 (1st). Capacity for 42 million.
- Airport of São Paulo – Congonhas (CGH).
19 million passengers in 2015 (3rd). Capacity for 17 million.
- International Airport of Viracopos (VCP).
10 million passengers in 2015 (6th). Capacity for 10 million.



Source: McKinsey & Company (2010)

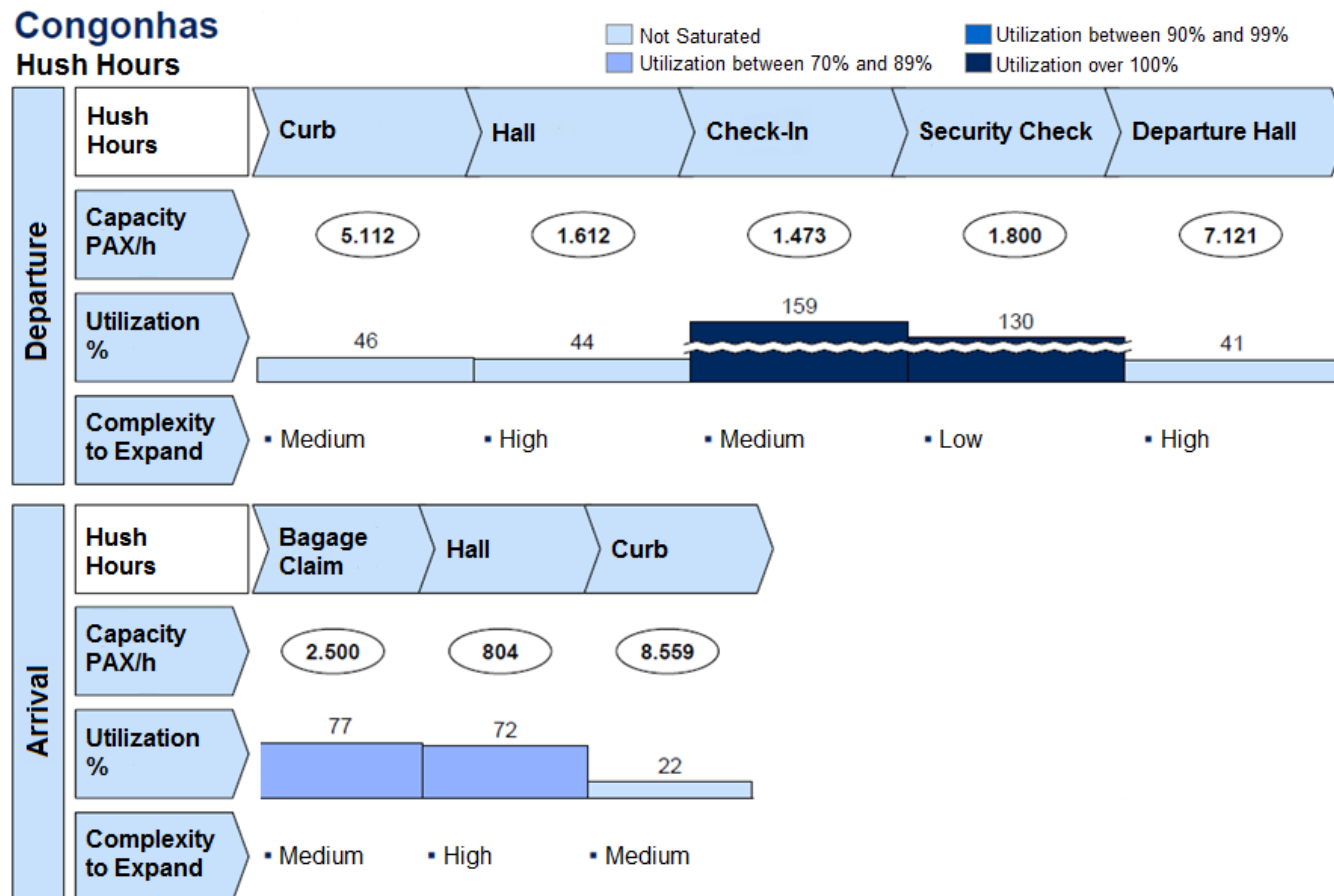
CONGONHAS AIRPORT

- 585 movements per day, on average.
- Number of passengers increased 58% from 2003 to 2015.
- Any growth in both runway systems and passenger terminals is infeasible.
- Congonhas have received almost no investments due to the FIFA World Cup 2014 and Rio 2016 Olympic Games.



CONGONHAS AIRPORT

- Check-in and security check areas in Congonhas Airport are fully saturated and operating with 159% and 130% utilization.



Source: McKinsey & Company (2010)

WHAT IS THE PROBLEM OF THIS SIMULATION STUDY?

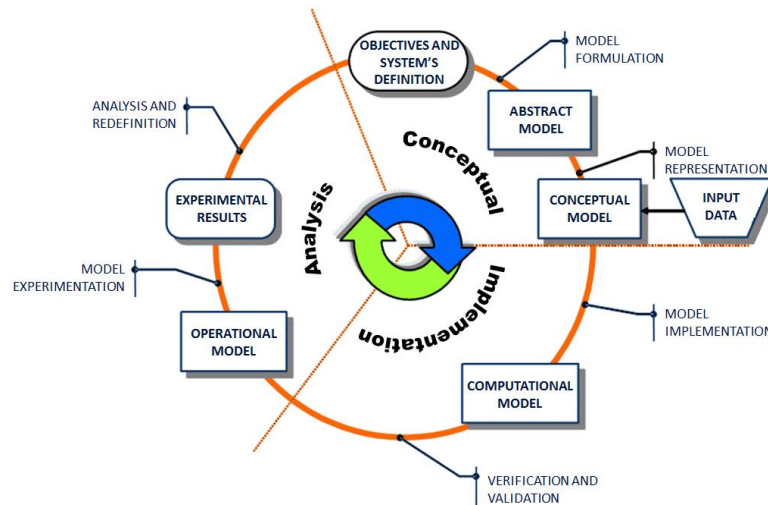
- Check-in at Congonhas Airport.
 - Congonhas has limited or no capacity for expansion.
 - Check-in is the bottleneck of this airport.
 - Customers spend on average 80% of the time in check-in.
 - The time spent on check-in directly impact customer's satisfaction.



CONCEPTUAL DESIGN

- Two major airlines, which represent 88% of the market share of Congonhas, are considered in this study. (Company A and Company B)
- Simulation project schedule was presented and approved by both airlines.
- Managers, supervisors, leaders and analysts were all involved. (Type III Error)
- Simulation Methodology: Chwif and Medina (2010)

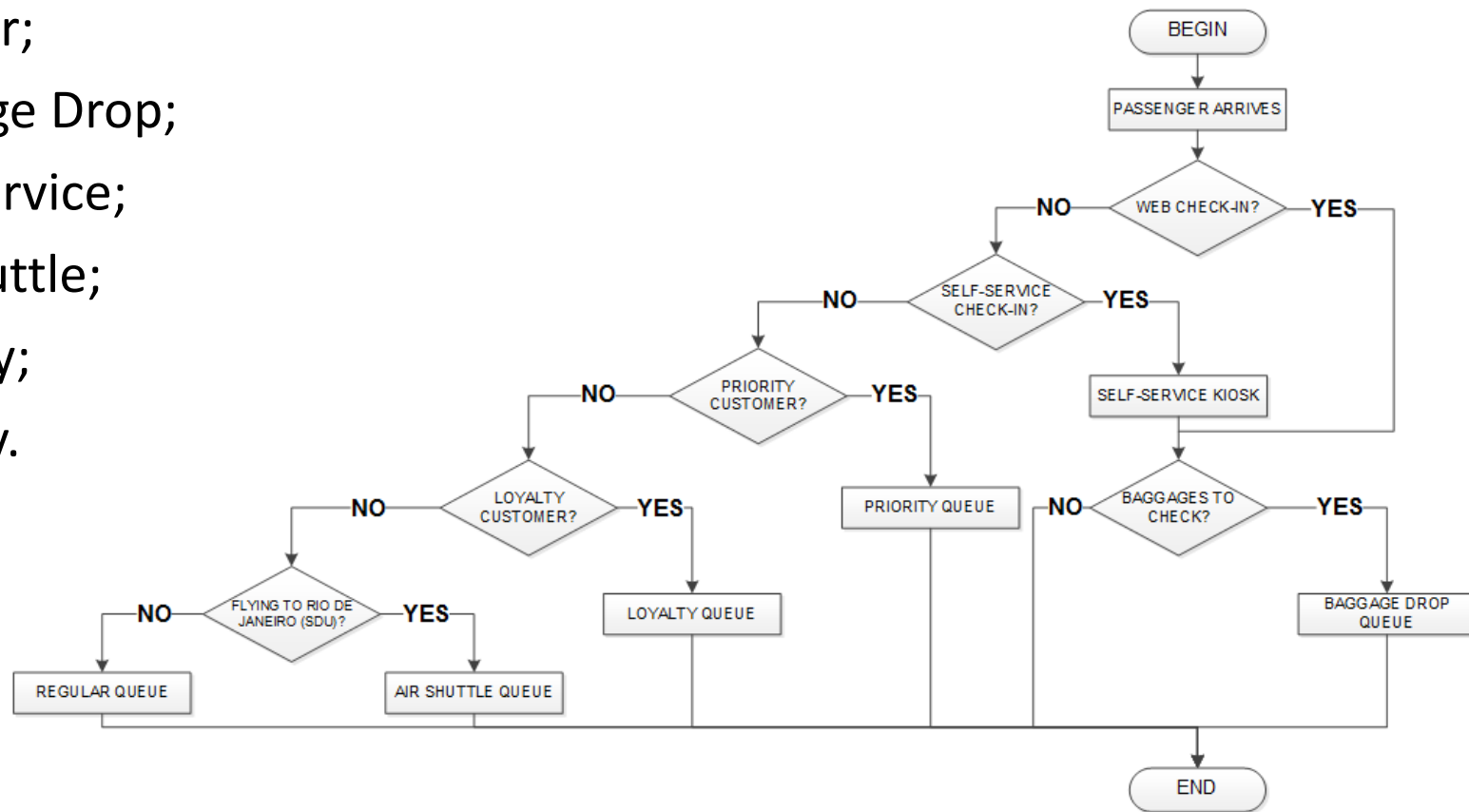
Banks and Chwif (2010)



Source: Chwif and Medina (2010)

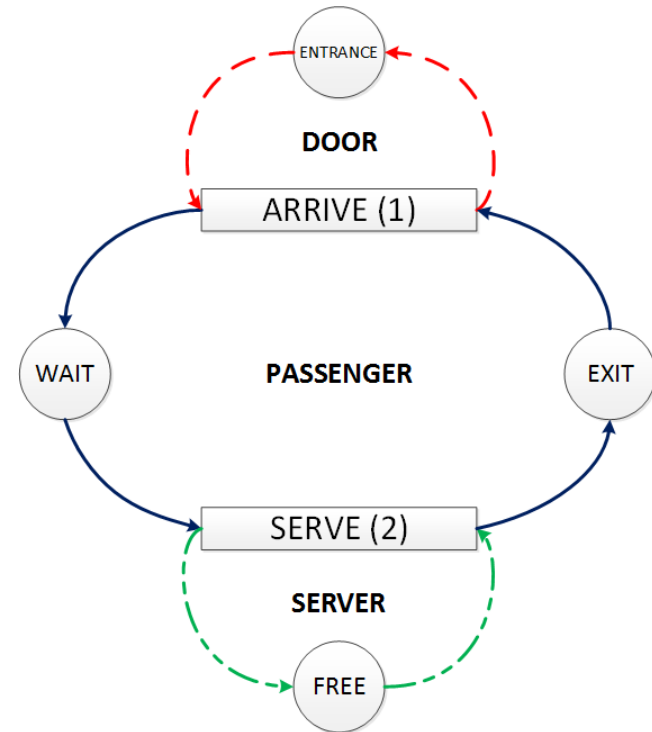
CONCEPTUAL MODEL

- Conceptual model built before the implementation of the computational model. Banks and Chwif (2010)
- Six classes of check-in were considered:
 - Regular;
 - Baggage Drop;
 - Self-Service;
 - Air Shuttle;
 - Priority;
 - Loyalty.

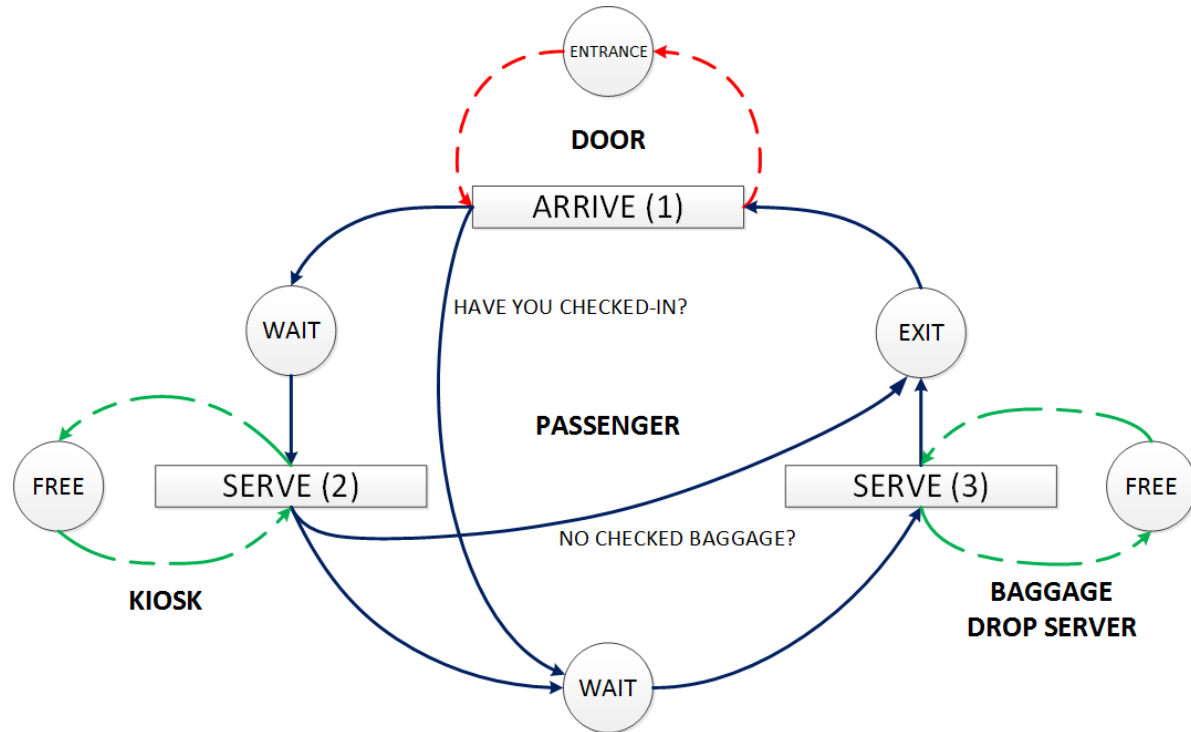


CONCEPTUAL MODEL

- Activity Cycle Diagram (ACD)



Regular, Air Shuttle,
Priority, and Loyalty



Baggage Drop and Self-Service

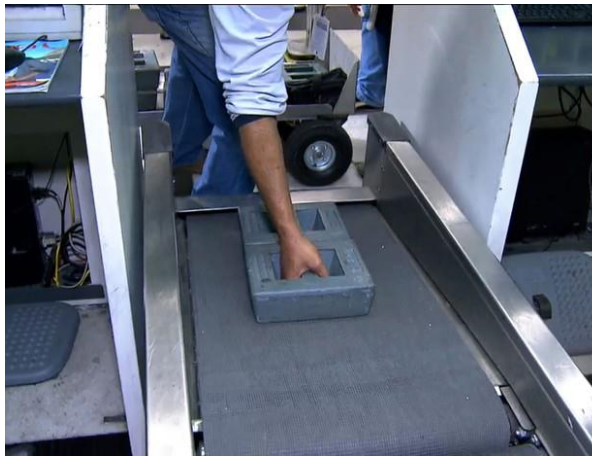
INPUT DATA AND OUTPUT RESULTS

- **Banks and Chwif (2010)** recommend defining input data properly:
 - Passenger arrival time before flight takeoff;
 - Service time for each class of check-in;
 - Employee's schedule for each class of check-in;
 - Load factor for scheduled flights;
 - % of customers served by each class of check-in;
 - % of customers who checked-in in groups;
 - Reason for travel, gender, age, travel destination, and suggestions.
- **Banks and Chwif (2010)** also suggest defining output data correctly:
 - % of customers waiting in line less than 12 minutes^{*};
 - % of customers waiting in line between 12 and 30 minutes^{*};
 - % of customers waiting in line more than 30 minutes^{*};
 - Average waiting time in queue for each class of check-in.

* IATA (2014). **Banks and Chwif (2010)** suggest defining appropriate performance measure for comparison.

ASSUMPTIONS

- Defining assumptions correctly is another warning determined by Banks and Chwif (2010):
 - This simulation project considered service delays due to shift change or breakdowns in baggage claim carousels*;
 - The service time acquired already considered employee's performance and passenger travel time between the queue and the counter.



* Breakdowns were initially ignored but primary results indicated that these factors significantly impact output results.

Banks and Chwif (2010)

DATA ACQUISITION

- Data was acquired from April to July.
- Three periods: morning, afternoon, and evening.
- Three demands: low, medium, and high.
- Data was also acquired during holidays and vacation periods.
- Data acquisition adhered the following process:
 - Sequence of random consecutive flights was chosen for each period;
 - Difference between takeoff times was approximately 4 hours;
 - Begin 2 hours before first flight and end 30 minutes before last flight;
 - Only data from passengers in this set was acquired.
- On average, 25 flights for each period.
- More than 220 hours of data.

Anticipate Problems with Input Data
Banks and Chwif (2010)

INPUT DATA ANALYSIS

- ANOVA (Analysis of Variance)
- Scatter of Input Data
- Autocorrelation of Input Data

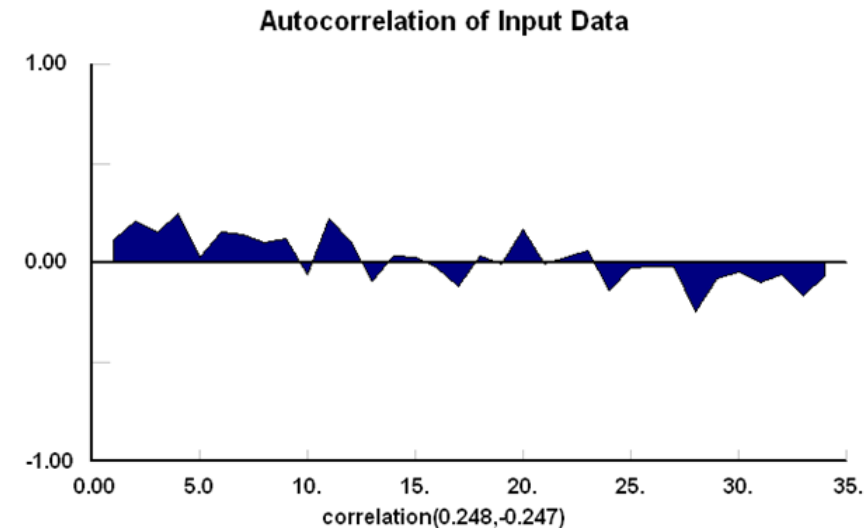
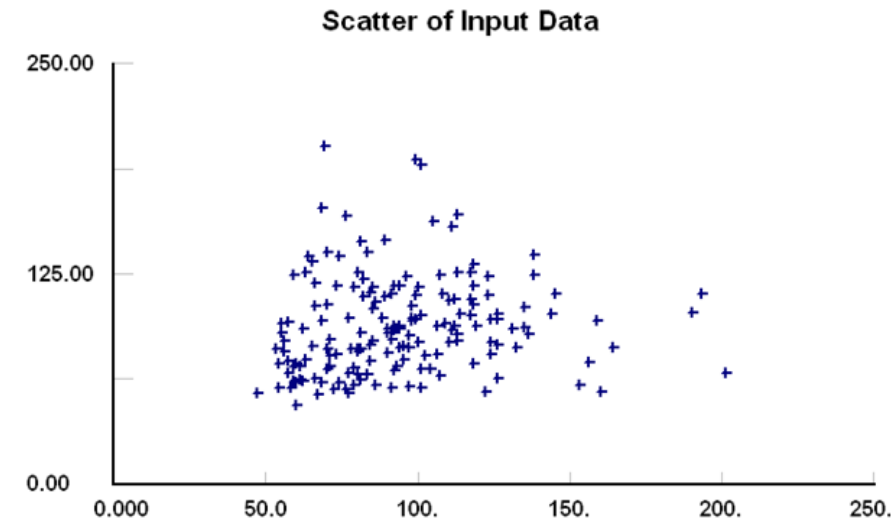
One-way ANOVA: Low, Medium, High

Source	DF	SS	MS	F	P
Factor	2	35052	17526	20,17	0,000
Error	643	558672	869		
Total	645	593723			

S = 29,48 R-Sq = 5,90% R-Sq(adj) = 5,61%

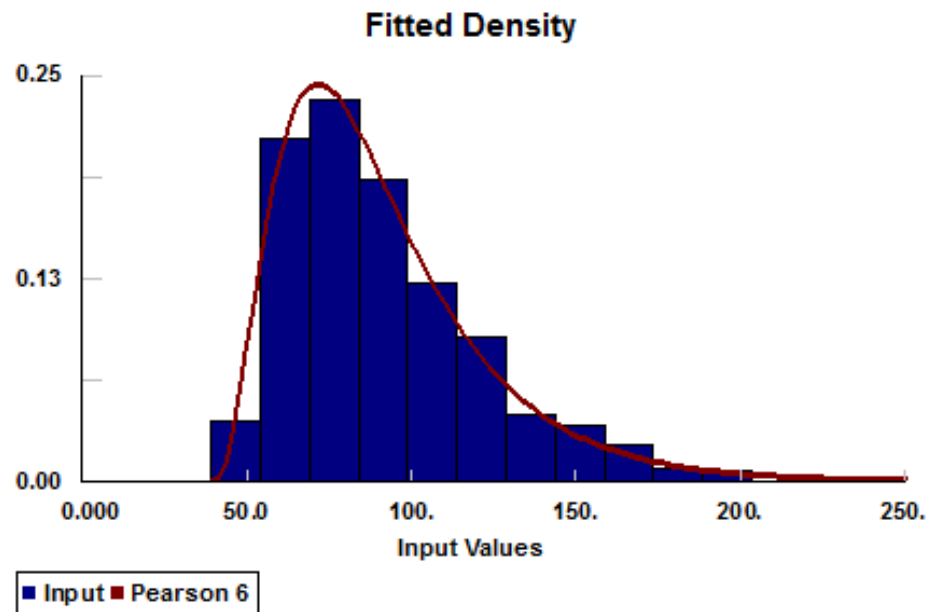
Individual 95% CIs For Mean Based on Pooled StDev			
Level	N	Mean	StDev
Low	141	103,51	35,41
Medium	172	93,87	28,64
High	333	85,08	27,05

Pooled StDev = 29,48



INPUT DATA ANALYSIS

- Fitted Density



Auto::Fit of Distributions

distribution	rank	acceptance
Pearson 6[43., 310, 2.14, 5.53]	92.8	do not reject
Lognormal[43., 4.65, 0.894]	85.1	do not reject
Gamma[43., 1.68, 86.2]	48.9	do not reject
Weibull[43., 1.29, 160]	35.3	do not reject
Beta[43., 1.06e+003, 1.35, 7.85]	24.1	do not reject
Exponential[43., 145]	6.22	do not reject
Pearson 5[43., 1.24, 82.]	1.79	do not reject
Erlang[43., 2., 86.2]	5.84e-002	reject
Triangular[42., 668, 42.]	8.66e-005	reject
Uniform[43., 650]	0.	reject
Rayleigh[43., 135]	0.	reject
Chi Squared[43., 106]	0.	reject
Power Function[43., 731, 0.531]	0.	reject

Pearson 6

minimum	=	43. [fixed]
beta	=	310.467
p	=	2.13939
q	=	5.53395

Kolmogorov-Smirnov

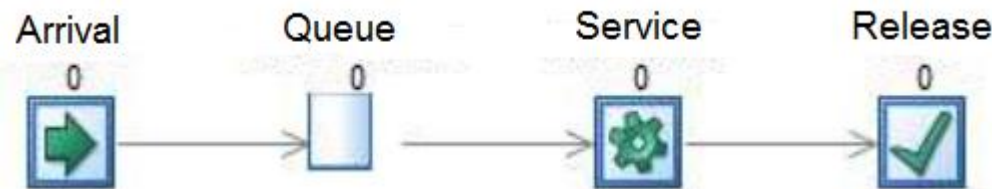
data points	54
ks stat	8.81e-002
alpha	5.e-002
ks stat[54,5.e-002]	0.181
p-value	0.763
result	DO NOT REJECT

Anderson-Darling

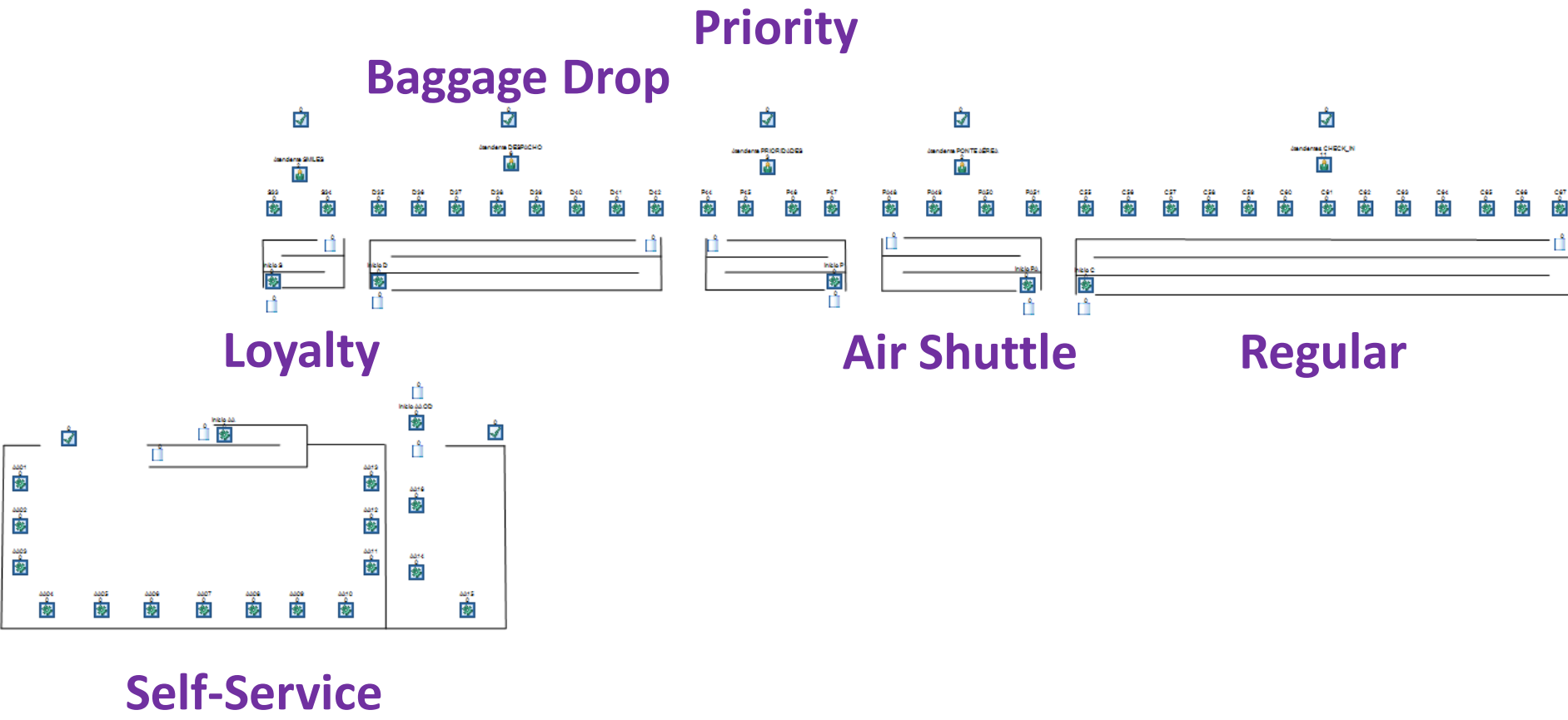
data points	53
ad stat	0.416
alpha	5.e-002
ad stat[5.e-002]	2.49
p-value	0.832
result	DO NOT REJECT

IMPLEMENTATION DESIGN

- The computational model created was complex enough to represent the system.
- Computational model was implemented gradually.
- Each class of check-in was implemented and validated separately prior to merging all models. [Banks and Chwif \(2010\)](#)
- Simulation Software: **SIMUL8**



COMPUTATIONAL MODEL



COMPUTATIONAL MODEL

Employees

Atendentes Conv

5



0



Release

Conv 1

0



Conv 2

0



Conv 3

0



Conv 4

0



Conv 5

0



Counters

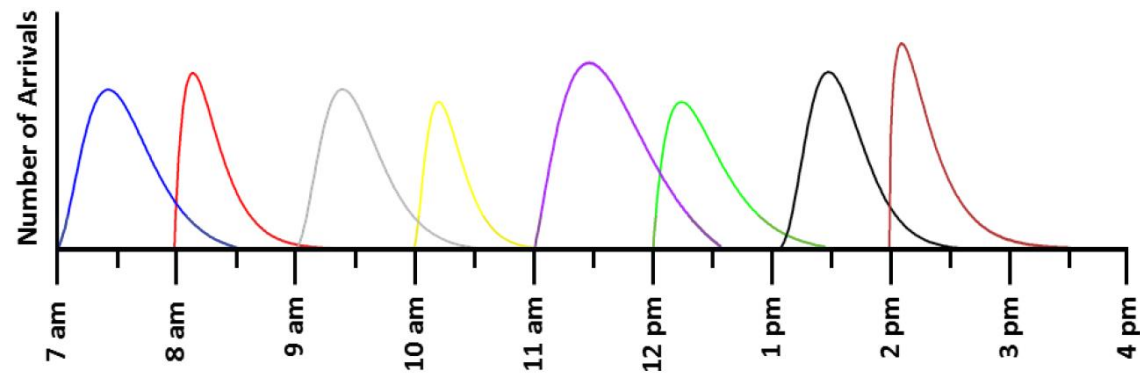


Queue

1389

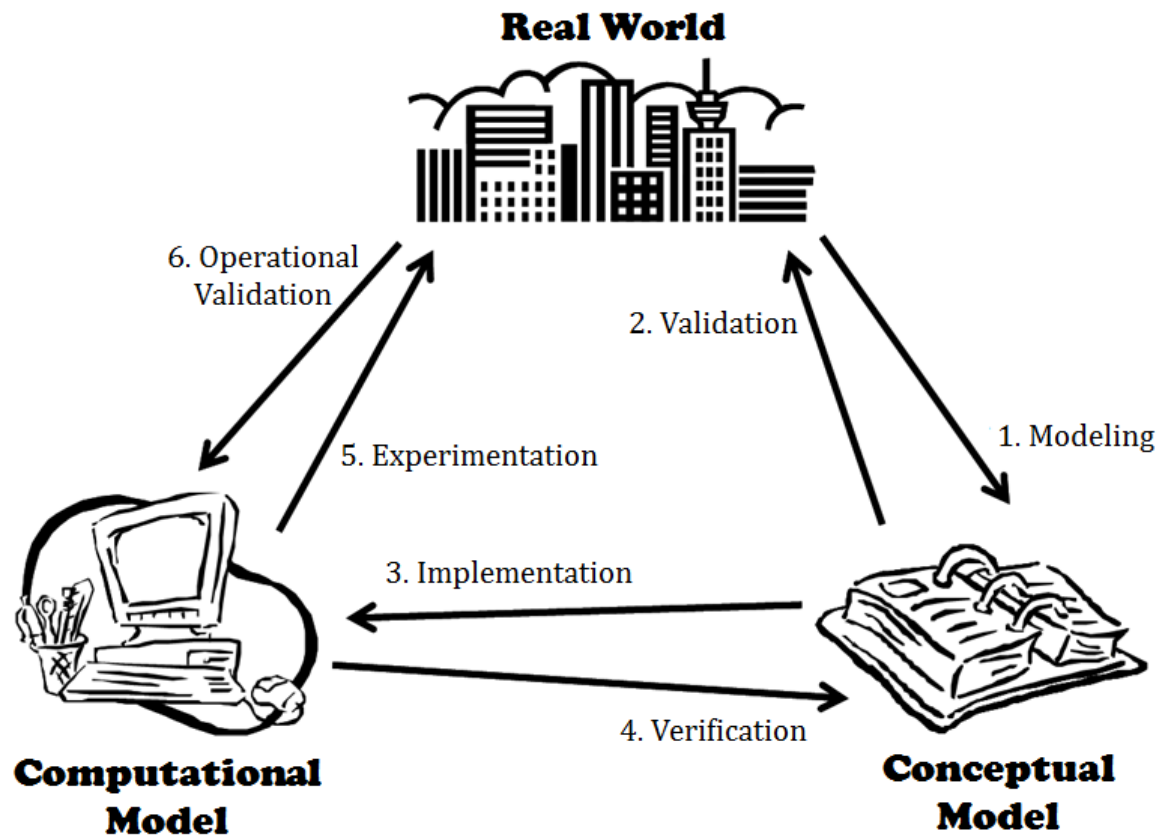


Arrival



BASE MODEL'S VERIFICATION

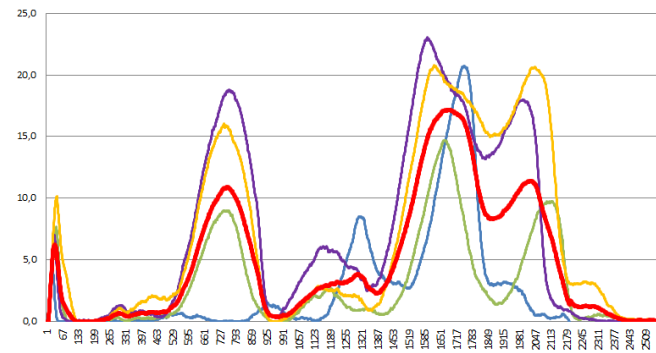
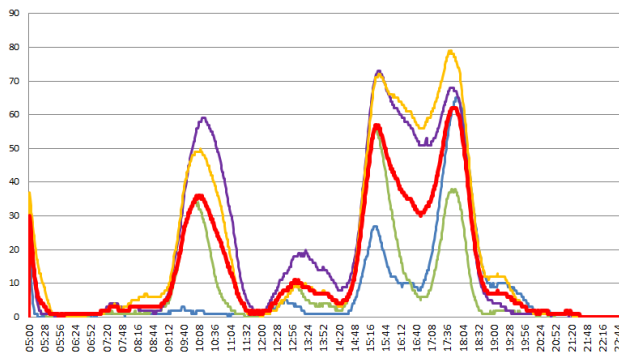
- Banks and Chwif (2010): “Do a lot of verification and validation, not a little”.
- Three techniques were used:
 - Modular Verification;
 - Graphical Animation;
 - Team Revision.



Source: Chwif and Medina (2010)

BASE MODEL'S VALIDATION

- Face to Face:
 - Output results were presented to managers, supervisors, leaders, and analysts of both airlines;
 - Three meetings were necessary with each airline before the actual validation of the computational model;
 - **Company A:** Increase / Decrease the number of passengers arriving in regular, air shuttle, and priority check-in;
 - **Company B:** Decrease the number of passengers arriving in priority check-in. Increase the number of resources in baggage drop queue.



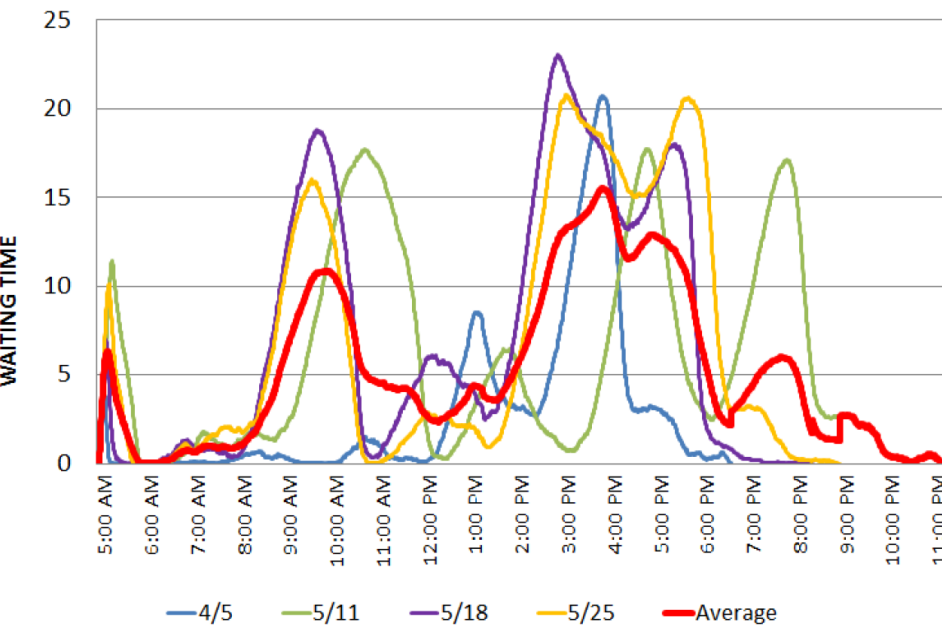
ANALYSIS DESIGN

- No warm-up period since Congonhas is a terminating system.
- This study considered:
 - 18 hours (5:00 am to 11:00 pm);
 - 7 days / week (Monday to Sunday);
 - Three shifts (5-11 am, 11 am to 5 pm, and 5-11 pm)
- 100 replications.
- Performance measures:
 - Less than 12 minutes in queue (Desired);
 - Between 12 and 30 minutes in queue (Too Long);
 - More than 30 minutes in queue (Unacceptable).
- All recommendations from [Banks and Chwif \(2010\)](#).



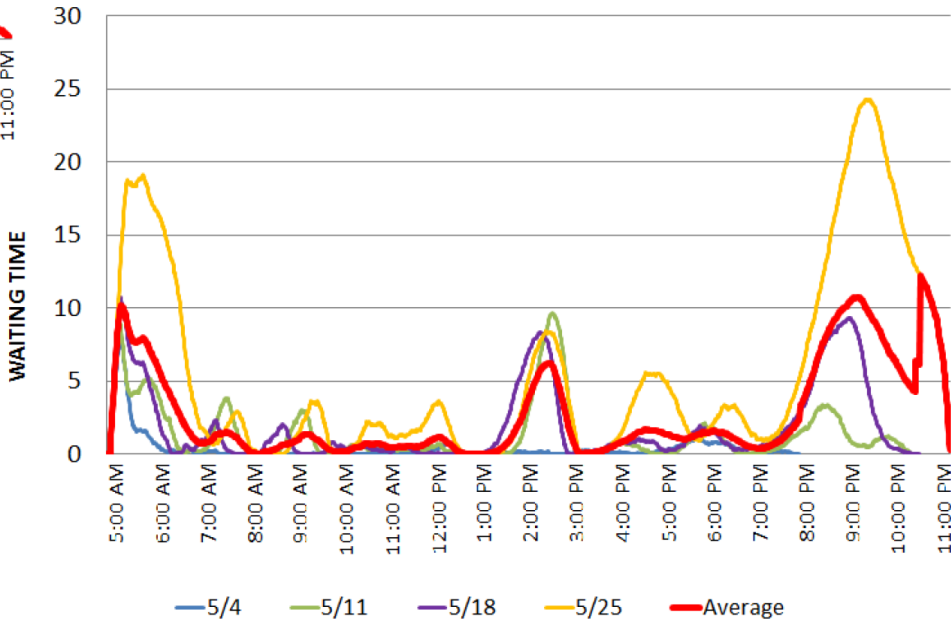
BASE MODEL RESULTS

- Current Demand Regular Check-In



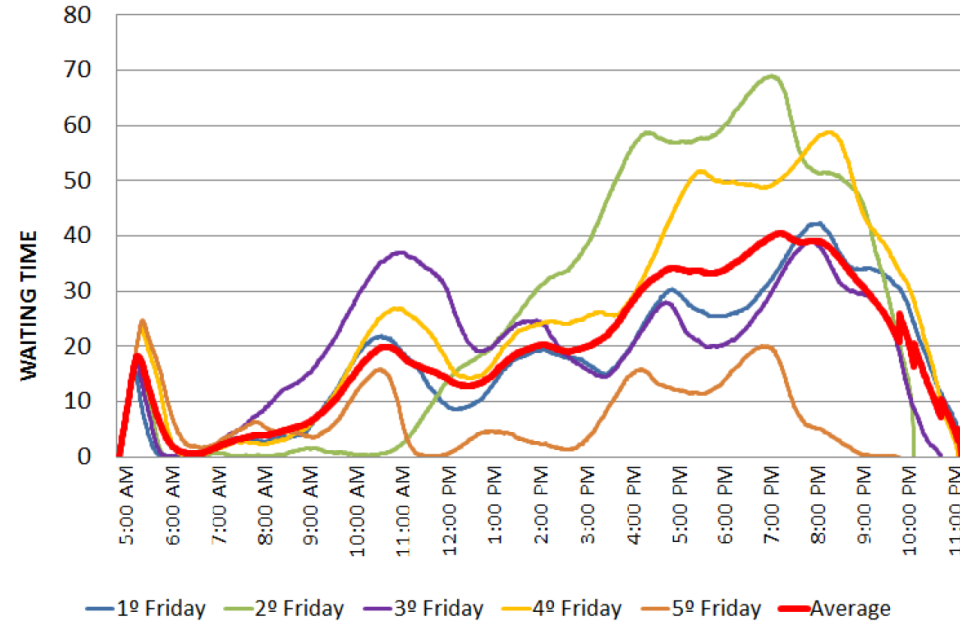
Company A

Company B



BASE MODEL RESULTS

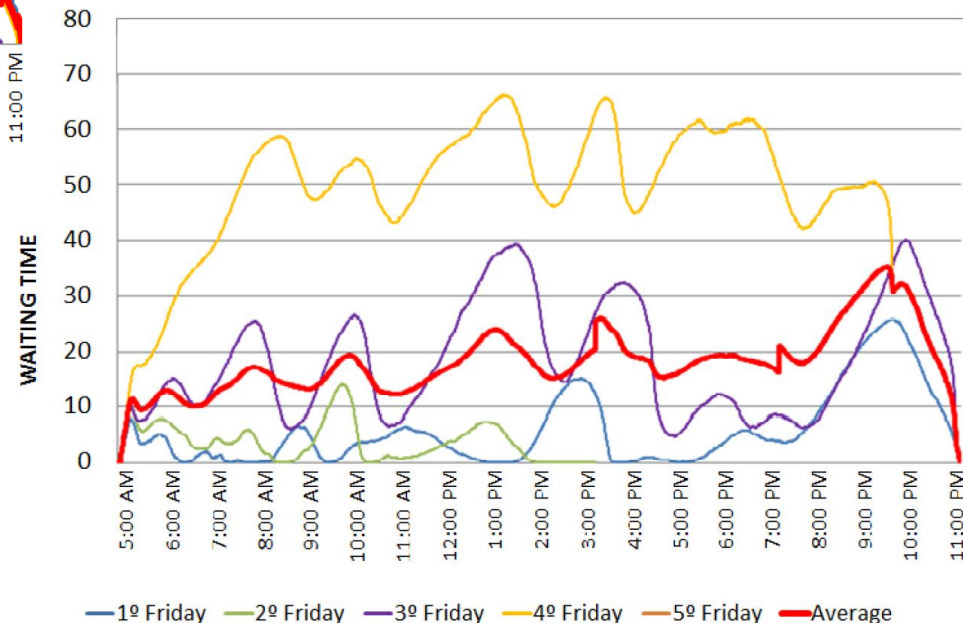
- Future Demand Regular Check-In



Company A

- Demand increases 4.2% per year.
- 100% load factor is an upper bound.

Company B



BASE MODEL RESULTS

- % of passengers waiting in line for current and future demands.

		Current Demand			Future Demand		
		< 12	12 – 30	> 30	< 12	12 – 30	> 30
Company A	Regular	81.7%	18.3%	0.0%	38.3%	38.7%	23.0%
	Baggage Drop	89.8%	10.2%	0.0%	56.9%	35.1%	8.0%
	Self-Service	99.0%	1.0%	0.0%	51.5%	23.3%	25.1%
	Air Shuttle	89.3%	10.7%	0.0%	39.7%	16.8%	43.5%
	Priority	90.1%	9.4%	0.5%	84.3%	15.7%	0.0%
	Loyalty	76.4%	23.6%	0.0%	57.6%	36.9%	5.4%
Company B	Regular	95.1%	4.9%	0.0%	58.8%	17.2%	24.0%
	Baggage Drop	99.0%	1.0%	0.0%	57.4%	21.1%	21.5%
	Self-Service	100.0%	0.0%	0.0%	100.0%	0.0%	0.0%
	Air Shuttle	99.6%	0.4%	0.0%	96.9%	3.1%	0.0%
	Priority	100.0%	0.0%	0.0%	56.9%	21.1%	22.0%
	Loyalty	99.5%	0.5%	0.0%	96.8%	3.2%	0.0%

IMPROVEMENT SCENARIOS

- **Scenario 1:** Addition of counters.

Company A	Company B
98% < 12 minutes <u>+ Counters</u> - Regular - Baggage Drop - Air Shuttle - Loyalty <u>+ Kiosks</u>	99% < 12 minutes <u>+ Counters</u> - Regular - Baggage Drop

IMPROVEMENT SCENARIOS

- **Scenario 2:** Split of regular and baggage drop queues based on flight destination.

Company A	Company B
Majority > 12 minutes <u>Future Demand</u> - Regular 1: 77% > 12 minutes - Regular 2: 37% > 12 minutes - Baggage Drop 1: 85% > 12 minutes - Baggage Drop 2: 49% > 12 minutes	~20% > 12 minutes <u>Future Demand</u> - Regular 1: 27% > 12 minutes - Regular 2: 5% > 12 minutes - Baggage Drop 1: 23% > 12 minutes - Baggage Drop 2: 28% > 12 minutes

IMPROVEMENT SCENARIOS

- **Scenario 3:** Increase use of web and self-service check-in.

Company A	Company B
Partially Good <u>Increase 10% to 20%</u> - Regular: 97% < 12 minutes - Priority: 99% < 12 minutes - Loyalty: 96% < 12 minutes - Self Service: 28% > 12 minutes - Air Shuttle: 26% > 12 minutes - Baggage Drop: 78% > 12 minutes	Partially Good <u>Increase 10% to 20%</u> - Regular: 94% < 12 minutes - Self Service: 100% < 12 minutes - Air Shuttle: 99% < 12 minutes - Loyalty: 99% < 12 minutes - Priority: 29% > 12 minutes - Baggage Drop: 51% > 12 minutes

IMPROVEMENT SCENARIOS

- **Scenario 4:** Merging of regular and baggage drop queues.

Company A	Company B
Majority > 12 minutes <u>Future Demand</u> - On average 7% worst	Majority > 12 minutes <u>Future Demand</u> - On average 15% worst

IMPROVEMENT SCENARIOS

- **Scenario 5:** Assign all passengers with no baggage to check from all queues to the self-service check-in.

Company A	Company B
Partially Good <u>Future Demand</u> - Regular: 98% < 12 minutes - Priority: 100% < 12 minutes - Loyalty: 100% < 12 minutes - Self Service: 34% > 12 minutes - Air Shuttle: 16% > 12 minutes - Baggage Drop: 65% > 12 minutes	Partially Good <u>Future Demand</u> - Regular: 100% < 12 minutes - Self Service: 100% < 12 minutes - Air Shuttle: 98% < 12 minutes - Loyalty: 93% < 12 minutes - Baggage Drop: 12% > 12 minutes - Priority: 14% > 12 minutes

IMPROVEMENT SCENARIOS

- **Scenario 6:** New queue for customers with excess baggage.

Company A	Company B
Majority < 12 minutes <u>Future Demand</u> - New Queue: 90% < 12 minutes	Majority < 12 minutes <u>Future Demand</u> - New Queue: 99% < 12 minutes

IMPROVEMENT SCENARIOS

- **Scenario 7:** Split of regular and baggage drop queues based on the number of baggage per passenger.

Company A	Company B
Majority > 12 minutes <u>Future Demand</u> - Regular 1: 52 > 12 minutes - Regular 2: 50% > 12 minutes - Baggage Drop 1: 53% > 12 minutes - Baggage Drop 2: 80% > 12 minutes	Majority < 12 minutes <u>Future Demand</u> - Regular 1: 97% < 12 minutes - Regular 2: 90% < 12 minutes - Baggage Drop 1: 90% < 12 minutes - Baggage Drop 2: 98% < 12 minutes

IMPROVEMENT SCENARIOS

- **Scenario 8:** Use of tablets in regular check-in queue.

Company A

Majority < 12 minutes

Require Investment



Company B

Majority < 12 minutes

Require Investment



IMPROVEMENT SCENARIOS

- **Scenario 9:** Assign all passengers with baggage to check from self-service check-in to regular, air shuttle, priority, and loyalty + Scenario 5.

Company A	Company B
Majority < 12 minutes <u>Increase Labor by 16%</u>	Majority < 12 minutes <u>Increase Labor by 14%</u> <u>+ Counters</u> - Regular

IMPROVEMENT SCENARIOS

- **Scenario 10:** Combination of Scenarios 1 and 3.

Company A	Company B
Majority < 12 minutes <u>Increase Labor by 12%</u> <u>+ 3 Kiosks</u>	Majority < 12 minutes <u>+ Counters</u> - Regular: 3 Counters - Baggage Drop: 2 Counters

IMPROVEMENT SCENARIOS

- **Scenario 11:** Combination of Scenarios 1, 3, and 6.

Company A	Company B
Majority < 12 minutes <u>+ Counters</u> - New Queue: 7 Counters <u>+ 3 Kiosks</u>	Majority < 12 minutes <u>Increase Labor by 14%</u> <u>+ Counters</u> - New Queue: 3 Counters - Regular: 1 Counter - Baggage Drop: 2 Counters

IMPROVEMENT SCENARIOS

- **Scenario 12:** Combination of Scenarios 1, 3, 5, and 8.

Company A	Company B
Majority < 12 minutes <u>Increase Labor by 8%</u> <u>+ 6 Kiosks</u>	Majority < 12 minutes <u>Increase Labor by 14%</u> <u>+ Counters</u> - Baggage Drop: 2 Counters

RECOMMENDATIONS

- Select scenario(s) with the highest benefit cost ratio.

Company A	Company B
<u>Scenario 12:</u> - Use of tablets in regular check-in; - Increase use of web and self-service check-in by 10% to 20%; - Assign all passengers with no baggage to check from all queues to the self-service check-in; - Increase Labor by 8% - 6 Kiosks	<u>Scenario 12:</u> - Use of tablets in regular check-in; - Increase use of web and self-service check-in by 10% to 20%; - Assign all passengers with no baggage to check from all queues to the self-service check-in; - Increase Labor by 14% - 2 Counters in Baggage Drop

CONCLUSIONS

- Warnings about simulation were considered to create a simulation study from conceptual to analysis design.
- Important warnings in this study:
 - Frequent interaction with employees during the study;
 - Statistical distributions for arrival and service times;
 - Delays due to shift change or breakdowns in baggage claim carousels;
 - Forecast future demand with upper bounds.

FUTURE RESEARCH

- Expand the methodology of this study to other areas of Congonhas such as:
 - Departure and arrival halls;
 - Baggage claim areas;
 - Runways and taxiways.
- Incorporate events not considered in this study:
 - Cancelled flights;
 - Charter flights.
- Determine optimal schedule for employees.

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